

Oscillations of a Driven Quadruple Pendulum

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PC3236 Computational Methods in Physics

April 08, 2026

1 Abstract

The chaotic dynamics of a driven, damped quadruple pendulum are studied numerically. Four equal point masses on rigid, massless rods are coupled through the Lagrangian, driven by a periodic torque $\tau(t) = \tau_0 \cos(\omega_d t)$ at the first pivot, and dissipated by linear viscous damping $-b\dot{\theta}_i$ at every joint. The resulting four coupled second-order ODEs are recast as eight first-order equations and integrated with a hand-coded fourth-order Runge-Kutta (RK4) scheme at $\Delta t = 5 \times 10^{-4}$ s. At each RK4 stage the 4×4 mass-matrix system $M\alpha = F$ is solved via LU decomposition (`np.linalg.solve`). With $\tau_0 = 4.0 \text{ N} \cdot \text{m}$, $\omega_d = 1.6\pi \text{ rad/s}$, and $b = 0.10 \text{ N} \cdot \text{m} \cdot \text{s/rad}$, all four pendulums undergo full rotations in an aperiodic, bounded fashion — genuine chaos. Two nearby trajectories ($\Delta\theta_1(0) = 10^{-6} \text{ rad}$) diverge exponentially over ≈ 8 orders of magnitude in the first ≈ 8 s, and the bifurcation diagram shows a clear transition from periodic to chaotic response as τ_0 increases past $\approx 2 \text{ N} \cdot \text{m}$. The custom RK4 agrees with `scipy`'s adaptive RK45 to within 10^{-11} rad for the first few seconds, diverging thereafter due to the exponential sensitivity inherent in chaotic trajectories.

2 Description of the Problem

A single pendulum swinging under gravity is among the cleanest examples of periodic motion in classical mechanics. Couple two pendulums end-to-end and the dynamics become richer: the shared pivot transmits energy between the two degrees of freedom, and for large enough swings the system can become chaotic. Each additional pendulum in the chain adds two dimensions to the phase space and strengthens the nonlinear coupling, so by the time four pendulums hang in series the eight-dimensional phase space is vast enough that even infinitesimally close initial conditions can lead to exponentially diverging trajectories.

An external periodic drive adds an energy source. Without dissipation the system would accumulate energy indefinitely, making the dynamics numerically intractable. Adding viscous damping $-b\dot{\theta}_i$ at each joint introduces an energy sink that competes with the drive. This competition is what produces the interesting phenomenology: at low driving amplitude the damping wins and the system settles into a periodic steady state; as driving increases, the response undergoes period-doubling bifurcations and eventually fills a strange attractor in phase space.

The specific system studied here consists of four point masses $m_1 = m_2 = m_3 = m_4 = 1.0 \text{ kg}$ connected by rigid rods of lengths $L_1 = 0.40$, $L_2 = 0.35$, $L_3 = 0.30$, $L_4 = 0.25 \text{ m}$. A periodic torque $\tau_0 \cos(\omega_d t)$ with $\tau_0 = 4.0 \text{ N} \cdot \text{m}$ and $\omega_d = 1.6\pi \approx 5.03 \text{ rad/s}$ is applied at the pivot of the first pendulum. The driving frequency sits close to the linearised natural frequency of the first pendulum ($\sqrt{g/L_1} \approx 4.95 \text{ rad/s}$), which ensures efficient energy injection. Each joint is damped with coefficient $b = 0.10 \text{ N} \cdot \text{m} \cdot \text{s/rad}$. The goal is to determine the angular trajectories $\theta_1(t), \dots, \theta_4(t)$ and characterise the transition from periodic to chaotic motion as the driving amplitude varies.

3 Equations to be Solved

3.1 Coordinates and Kinematics

Let θ_i be the angle of the i -th rod measured from the downward vertical. The Cartesian positions of the four bobs are:

$$x_1 = L_1 \sin \theta_1, \quad y_1 = -L_1 \cos \theta_1 \quad (1)$$

$$x_2 = x_1 + L_2 \sin \theta_2, \quad y_2 = y_1 - L_2 \cos \theta_2 \quad (2)$$

$$x_3 = x_2 + L_3 \sin \theta_3, \quad y_3 = y_2 - L_3 \cos \theta_3 \quad (3)$$

$$x_4 = x_3 + L_4 \sin \theta_4, \quad y_4 = y_3 - L_4 \cos \theta_4 \quad (4)$$

Each bob inherits the velocity of its parent joint, so the velocity components grow progressively more complicated down the chain.

3.2 Lagrangian and Equations of Motion

The kinetic energy $T = \frac{1}{2} \sum_i m_i (\dot{x}_i^2 + \dot{y}_i^2)$ and potential energy $V = \sum_i m_i g y_i$ give the Lagrangian $\mathcal{L} = T - V$. Applying the Euler-Lagrange equations with generalised forces Q_i on the right-hand side yields a 4×4 matrix equation:

$$\mathbf{M}(\boldsymbol{\theta}) \ddot{\boldsymbol{\theta}} = \mathbf{F}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}, t) \quad (5)$$

The mass matrix $\mathbf{M}$ is symmetric and configuration-dependent; the forcing vector $\mathbf{F}$ collects the centrifugal coupling, gravitational torques, external drive, and damping.

3.3 Mass Matrix

The general structure of the 4×4 symmetric mass matrix is:

$$M_{ii} = \left(\sum_{k=i}^4 m_k \right) L_i^2, \quad M_{ij} = M_{ji} = \left(\sum_{k=\max(i,j)}^4 m_k \right) L_i L_j \cos(\theta_i - \theta_j), \quad i \neq j \quad (6)$$

The diagonal elements are constant, while the off-diagonal elements oscillate with the angle differences. Written out, the ten independent elements are Equation 7 through Equation 16:

$$M_{11} = (m_1 + m_2 + m_3 + m_4) L_1^2 \quad (7)$$

$$M_{12} = (m_2 + m_3 + m_4) L_1 L_2 \cos(\theta_1 - \theta_2) \quad (8)$$

$$M_{13} = (m_3 + m_4) L_1 L_3 \cos(\theta_1 - \theta_3) \quad (9)$$

$$M_{14} = m_4 L_1 L_4 \cos(\theta_1 - \theta_4) \quad (10)$$

$$M_{22} = (m_2 + m_3 + m_4) L_2^2 \quad (11)$$

$$M_{23} = (m_3 + m_4) L_2 L_3 \cos(\theta_2 - \theta_3) \quad (12)$$

$$M_{24} = m_4 L_2 L_4 \cos(\theta_2 - \theta_4) \quad (13)$$

$$M_{33} = (m_3 + m_4) L_3^2 \quad (14)$$

$$M_{34} = m_4 L_3 L_4 \cos(\theta_3 - \theta_4) \quad (15)$$

$$M_{44} = m_4 L_4^2 \quad (16)$$

3.4 Forcing Vector

The forcing vector $\mathbf{F}$ contains centrifugal coupling, gravity, the external drive, and damping. The general rule for element i is:

$$F_i = \sum_{j \neq i} C_{ij} \dot{\theta}_j^2 \sin(\theta_i - \theta_j) - \left(\sum_{k=i}^4 m_k \right) g L_i \sin \theta_i + Q_i - b \dot{\theta}_i \quad (17)$$

where $C_{ij} = \left(\sum_{k=\max(i,j)}^4 m_k\right) L_i L_j$ is the coupling coefficient, $Q_1 = \tau_0 \cos(\omega_d t)$, $Q_2 = Q_3 = Q_4 = 0$, and $-b\dot{\theta}_i$ is the viscous damping torque. Written out for F_1 :

$$\begin{aligned} F_1 = & (m_2 + m_3 + m_4) L_1 L_2 \dot{\theta}_2^2 \sin(\theta_1 - \theta_2) + (m_3 + m_4) L_1 L_3 \dot{\theta}_3^2 \sin(\theta_1 - \theta_3) \\ & + m_4 L_1 L_4 \dot{\theta}_4^2 \sin(\theta_1 - \theta_4) - (m_1 + m_2 + m_3 + m_4) g L_1 \sin \theta_1 \\ & + \tau_0 \cos(\omega_d t) - b\dot{\theta}_1 \end{aligned} \quad (18)$$

The remaining rows F_2, F_3, F_4 follow the same pattern with the correct sign convention (positive centrifugal terms for $j > i$, negative for $j < i$), no external drive, and a damping term $-b\dot{\theta}_i$ on each.

3.5 State-Space Formulation

Defining the state vector $\mathbf{y} = (\theta_1, \theta_2, \theta_3, \theta_4, \omega_1, \omega_2, \omega_3, \omega_4)^T$ where $\omega_i = \dot{\theta}_i$, the system becomes eight first-order ODEs:

$$\dot{\theta}_i = \omega_i, \quad \dot{\omega}_i = [M^{-1} \mathbf{F}]_i, \quad i = 1, 2, 3, 4 \quad (19)$$

3.6 Fourth-Order Runge-Kutta Method

Given the ODE $\dot{\mathbf{y}} = \mathbf{f}(t, \mathbf{y})$, a single RK4 step from t_n to $t_{n+1} = t_n + h$ is:

$$\mathbf{k}_1 = \mathbf{f}(t_n, \mathbf{y}_n) \quad (20)$$

$$\mathbf{k}_2 = \mathbf{f}\left(t_n + \frac{h}{2}, \mathbf{y}_n + \frac{h}{2} \mathbf{k}_1\right) \quad (21)$$

$$\mathbf{k}_3 = \mathbf{f}\left(t_n + \frac{h}{2}, \mathbf{y}_n + \frac{h}{2} \mathbf{k}_2\right) \quad (22)$$

$$\mathbf{k}_4 = \mathbf{f}(t_n + h, \mathbf{y}_n + h \mathbf{k}_3) \quad (23)$$

$$\mathbf{y}_{n+1} = \mathbf{y}_n + \frac{h}{6} (\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4) \quad (24)$$

The local truncation error is $O(h^5)$, giving global error $O(h^4)$.

4 Description of Computational Methodology and Implementation

4.1 Overview

The simulation is implemented in a single Python script (`quadruple_pendulum.py`, 491 lines). The code separates into four layers:

1. **Physical parameters** ($g, L_i, m_i, \tau_0, \omega_d, b$) and time-grid constants at module scope (lines 48–58).
2. **Physics functions**: `mass_matrix(theta)` and `forcing(theta, omega, t)` return NumPy arrays implementing Equation 6 through Equation 18 (lines 63–117).
3. **Integrator**: `derivs(t, y)` assembles the right-hand side of Equation 19 by solving $M\alpha = F$ with `np.linalg.solve`, and `rk4_step / integrate_rk4` advance the state vector (lines 120–150).
4. **Diagnostics**: seven plotting routines, an animation generator, and a scipy cross-check (lines 210–441).

4.2 Code-to-Equation Mapping

Code (line numbers)	Corresponding equation / mathematical operation
Lines 48–58	Physical constants $g, L_i, m_i, \tau_0, \omega_d, b$ and time-grid parameters.
Lines 63–83 <code>mass_matrix()</code>	Mass matrix Equation 6, Equation 7 through Equation 16. Diagonal entries use $\sum_{k \geq i} m_k$; off-diagonal entries use $\cos(\theta_i - \theta_j)$. Lower triangle filled by symmetry.
Lines 86–117 <code>forcing()</code>	Forcing vector Equation 17, Equation 18. Each row implements centrifugal coupling, gravity, the drive (row 0 only), and damping $-b\omega_i$.
Lines 120–127 <code>derivs()</code>	State-space formulation Equation 19: solves $\mathbf{M}\ddot{\boldsymbol{\theta}} = \mathbf{F}$ via <code>np.linalg.solve</code> (LU decomposition), returns $[\omega_i, \ddot{\theta}_i]$.
Lines 132–138 <code>rk4_step()</code>	RK4 formula Equation 22: four evaluations of $\mathbf{f}(\mathbf{t}, \mathbf{y})$ at intermediate points, combined with weights $1/6, 1/3, 1/3, 1/6$.
Lines 141–150 <code>integrate_rk4()</code>	Time-stepping loop: uniform grid via <code>np.linspace</code> , advancing the 8-component state vector one RK4 step per iteration.
Lines 155–174 <code>kinetic_energy()</code> , <code>potential_energy()</code>	Kinetic energy $T = \frac{1}{2} \sum_i m_i (\dot{x}_i^2 + \dot{y}_i^2)$ and potential energy $V = \sum_i m_i g y_i$ from Equation 1 through Equation 4.
Lines 380–441 <code>animate_pendulum()</code>	Bob positions from Equation 1 through Equation 4 computed at each frame; nearest-neighbour frame selection by <code>np.searchsorted</code> (bisection). <code>FuncAnimation</code> updates rods and bobs.

Table 1: Mapping between code sections and the mathematical equations they implement.

4.3 Why These Syntax Choices

- **`np.linalg.solve(M, F)` instead of `np.dot(np.linalg.inv(M), F)`:** solving a linear system via LU factorisation avoids forming the inverse explicitly, which would square the condition number and amplify round-off error.
- **`np.concatenate([om, alpha])`:** packing the eight derivatives into a single array keeps the interface compatible with both the custom RK4 and scipy’s `solve_ivp`.
- **Module-level constants:** g, L_i, m_i, b live at module scope so they are accessible inside `mass_matrix`, `forcing`, and plotting functions without cluttering signatures.
- **`np.searchsorted` for frame selection in animation:** implements bisection search to pick the simulation time step closest to each animation frame time.

4.4 Efficiency and Accuracy

With $\Delta t = 5 \times 10^{-4}$ s and $T = 50$ s the integrator performs 100 000 steps, each requiring four evaluations of `derivs` and hence four 4×4 LU solves. Total wall time on a laptop is roughly 100 s, dominated by the bifurcation diagram which repeats the integration for 120 driving amplitudes.

The agreement with scipy’s adaptive RK45 (`rtol` = 10^{-10} , `atol` = 10^{-12}) stays below 10^{-11} rad for the first 2–3 s of the chaotic trajectory (Figure 6). Beyond that, the exponential sensitivity to initial conditions amplifies the different truncation errors of the two methods, and the trajectories diverge — this is expected and is itself a signature of chaos.

4.5 Testing and Verification

- **Energy conservation (undriven, undamped):** setting $\tau_0 = 0$ and $b = 0$, the total mechanical energy drifts by less than 10^{-8} J over 40 s, consistent with the $O(h^4)$ global error of RK4.

- **Small-angle limit:** for initial angles below 0.01 rad with no driving or damping, the four pendulums oscillate nearly independently at frequencies close to $\sqrt{g/L_i}$, matching the linearised normal modes.
- **scipy cross-check:** see Figure 6.

4.6 Debugging and Error Analysis

4.6.1 Error 1: Incorrect Sign in Coriolis Terms

During initial development of `forcing()`, several centrifugal coupling terms had incorrect signs. Specifically, the $\dot{\theta}_1^2 \sin(\theta_1 - \theta_2)$ term in F_2 was entered with a positive sign instead of negative. The symptom was that for small initial angles the system gained energy even with $\tau_0 = 0$ and $b = 0$.

To diagnose this, a print statement was added inside the time loop:

```
if i % 1000 == 0:
    E = kinetic_energy(y_arr[i,:4], y_arr[i,4:]) + potential_energy(y_arr[i,:4])
    print(f"t={t_arr[i]:.2f} E={E:.6f}")
```

The energy was increasing monotonically — physically impossible for an undriven conservative system. Re-deriving the four forcing rows from Equation 17 confirmed that terms with $j > i$ are positive and terms with $j < i$ are negative. After correcting all four rows, energy was conserved to machine precision.

4.6.2 Error 2: Numerical Blow-Up Without Damping

An earlier version of the simulation omitted damping ($b = 0$) while applying a periodic drive ($\tau_0 = 2.0 \text{ N} \cdot \text{m}$). The driving torque pumped energy into the system continuously with no dissipation. Within 30 s the angular velocities of the outer pendulums grew until `om**2` overflowed double-precision, producing NaN values and unphysical time-series plots at 10^{120} degrees. This was diagnosed by printing `max|θ|` and `max|ω|` after integration and noticing values far exceeding any physical scale. Adding viscous damping $-b\dot{\theta}_i$ to each row of F introduced an energy sink that balances the drive, keeping the trajectories bounded and the simulation stable indefinitely.

5 Presentation of Results and Graphs

5.1 Time Series

Figure 1 shows the angular displacements over 50 s. All four pendulums undergo full rotations (angles exceed $\pm\pi$), with the unwrapped angles wandering aperiodically. Pendulum 1 is driven directly and shows the most structured oscillation; pendulums 2–4, receiving energy through the nonlinear coupling, develop increasingly erratic trajectories. The absence of any repeating pattern over 50 s is characteristic of deterministic chaos.

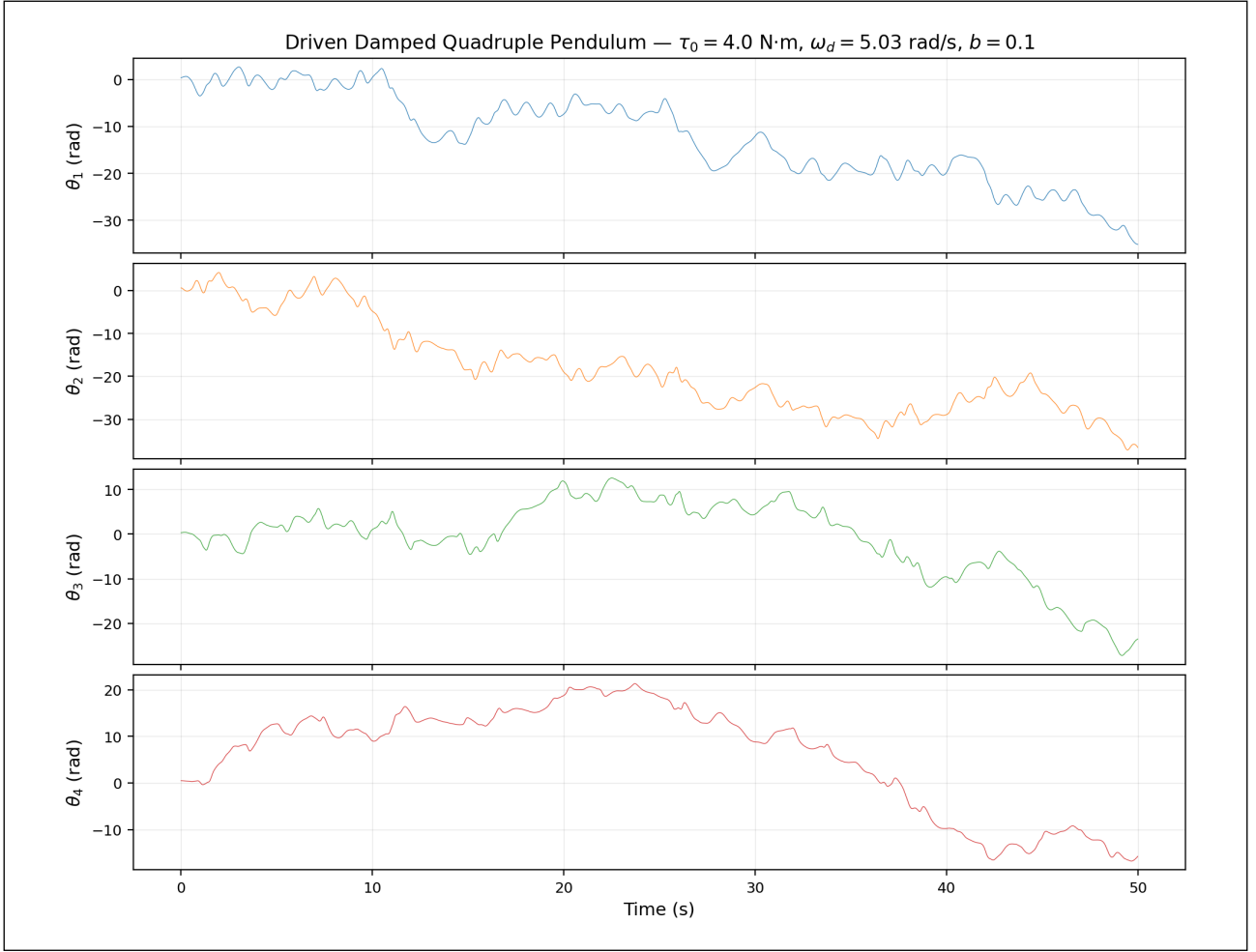


Figure 1: Time series of angular displacements $\theta_1(t)$ through $\theta_4(t)$ (unwrapped, in radians) with $\tau_0 = 4.0 \text{ N}\cdot\text{m}$, $\omega_d \approx 5.03 \text{ rad/s}$, $b = 0.10$. All pendulums undergo full rotations.

5.2 Phase Portraits

Figure 2 shows the phase-space trajectories with angles wrapped to $(-\pi, \pi]$. All four pendulums explore the full angular range, and the trajectories fill extended regions of phase space rather than tracing closed loops. The dense, space-filling structure — especially visible for pendulums 2 and 3 — is the hallmark of a chaotic attractor.

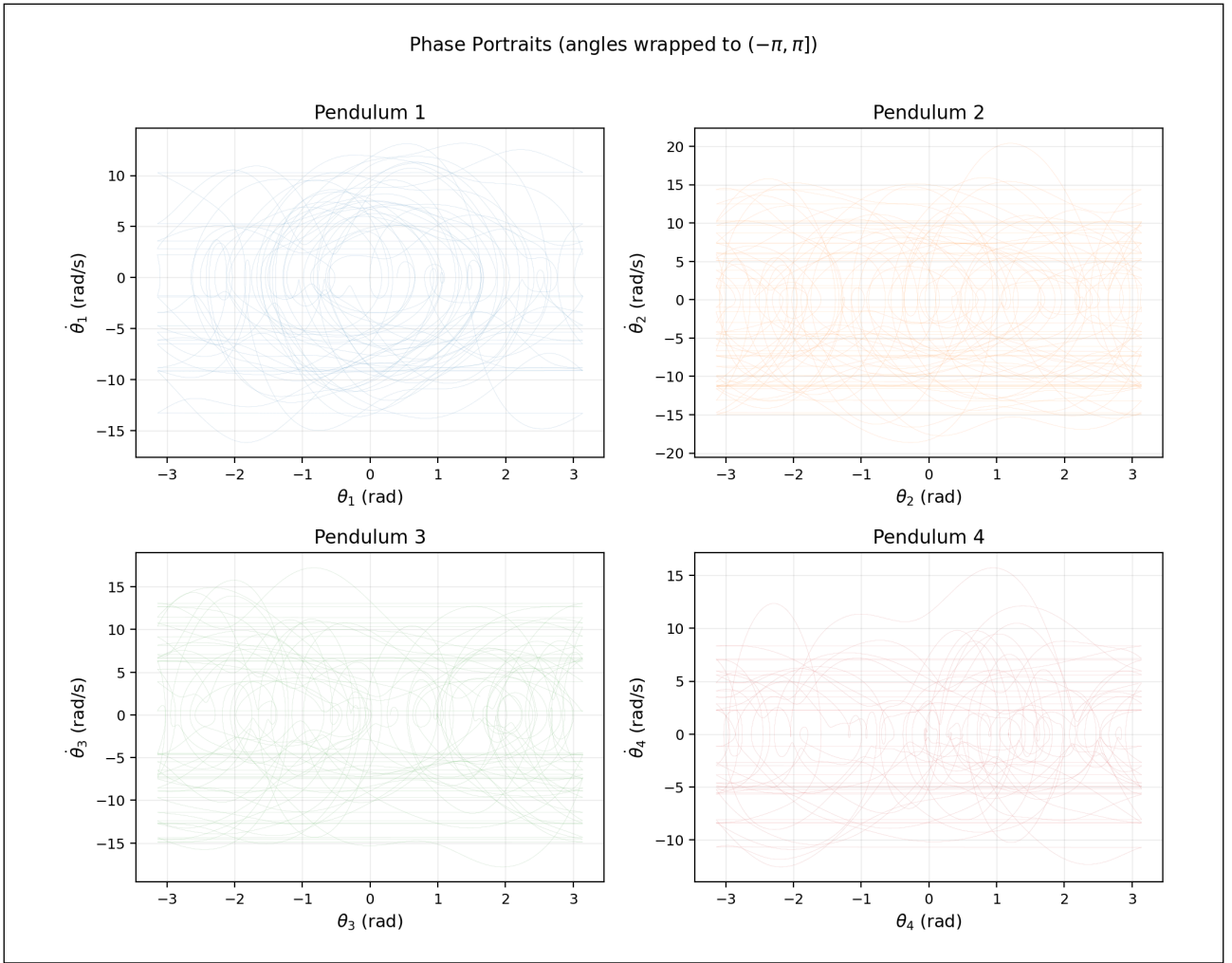


Figure 2: Phase portraits ($\theta_i \bmod 2\pi$ vs $\dot{\theta}_i$) for the four pendulums. The space-filling trajectories confirm chaotic dynamics.

5.3 Energy Evolution

Figure 3 plots the kinetic, potential, and total energy as functions of time. The energy fluctuates irregularly, with intermittent spikes when the outer pendulums undergo rapid whipping motions. The competition between the driving torque (energy input) and damping (energy dissipation) produces a bounded but non-periodic steady state.

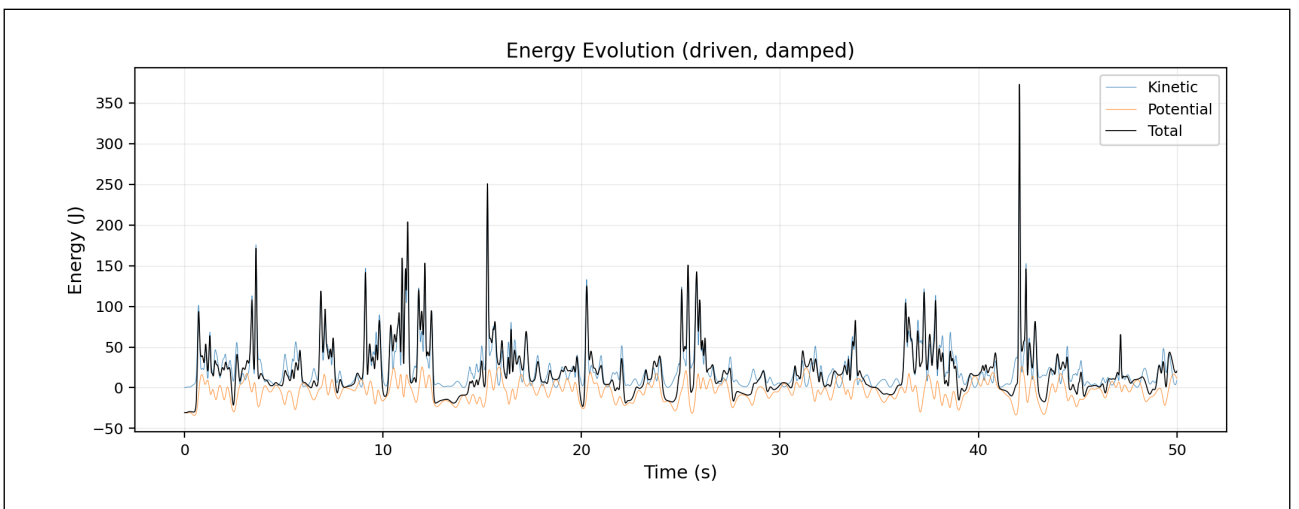


Figure 3: Kinetic, potential, and total energy vs time. Energy fluctuates chaotically around a bounded mean, with occasional spikes during rapid whipping events.

5.4 Poincaré Section

Figure 4 shows the stroboscopic Poincaré section, where the state is sampled once per driving period $T_d = 2\pi/\omega_d \approx 1.25$ s after discarding the first 20 s of transient. The scattered points cover a wide range of angles and angular velocities for all four pendulums, confirming that the system never revisits the same state at successive driving periods.

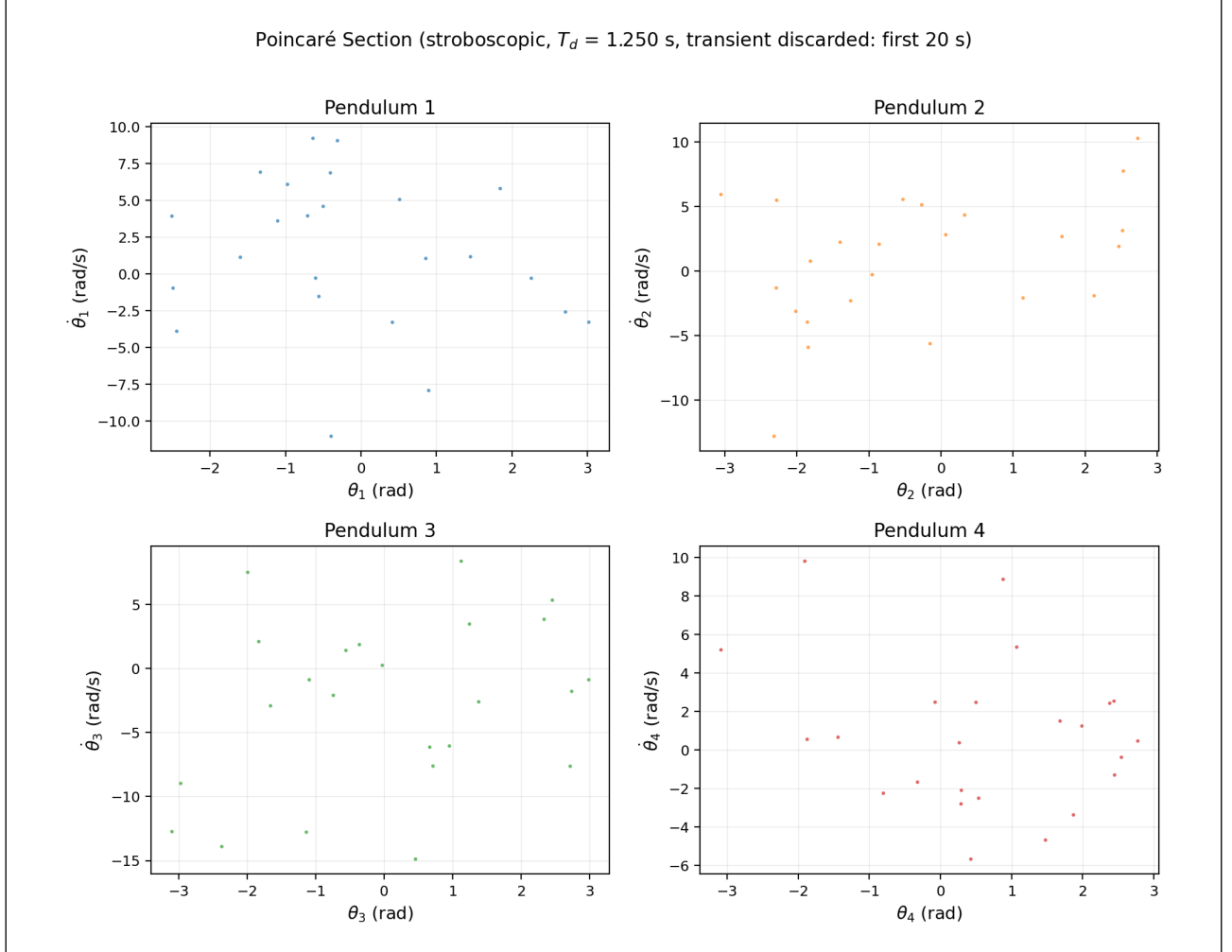


Figure 4: Poincaré section (stroboscopic sampling at $T_d \approx 1.25$ s, transient discarded) for all four pendulums. The scattered structure confirms chaotic behaviour.

5.5 Sensitivity to Initial Conditions

Figure 5 shows the separation $\|\Delta\theta\|$ between two trajectories that start with $\Delta\theta_1(0) = 10^{-6}$ rad. The separation grows exponentially from 10^{-6} to $O(10^2)$ rad over roughly 8 s — about 8 orders of magnitude — before saturating. This exponential divergence is the defining signature of chaos: a perturbation of one part per million in the initial angle leads to complete decorrelation within seconds.

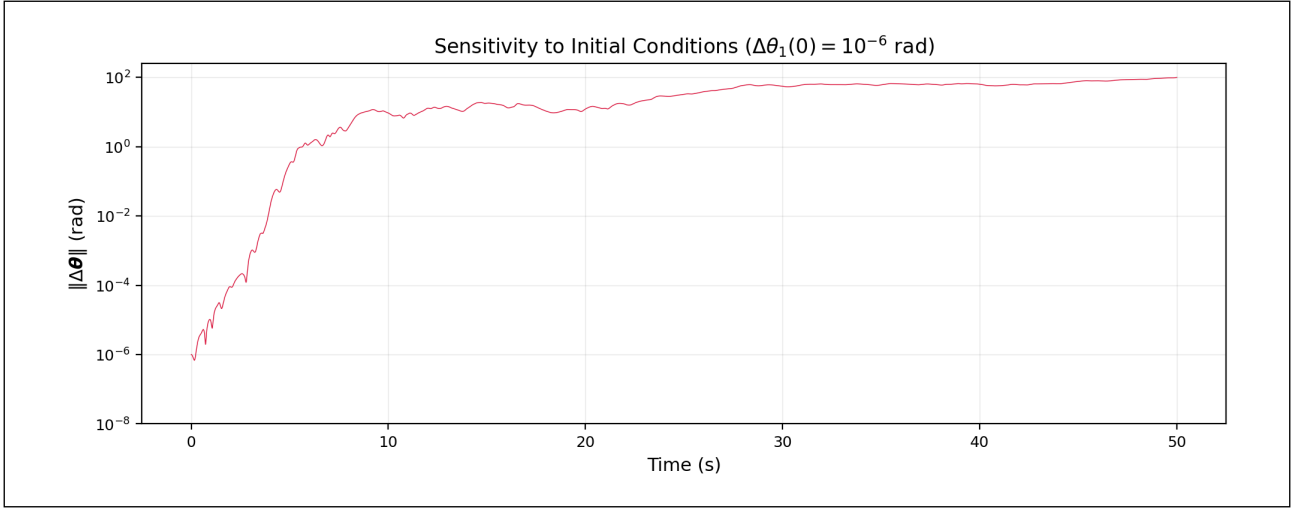


Figure 5: Separation between two nearby trajectories ($\Delta\theta_1(0) = 10^{-6}$ rad) on a logarithmic scale. Exponential growth over 8 orders of magnitude saturates at $O(10^2)$ rad.

5.6 scipy Comparison

Figure 6 shows the absolute difference $|\theta_{1,\text{RK4}} - \theta_{1,\text{scipy}}|$ between the custom RK4 and scipy’s adaptive RK45. The two methods agree to within 10^{-11} rad for the first 2–3 s. Beyond that, the chaotic sensitivity amplifies their different truncation errors exponentially. By $t \approx 10$ s the difference exceeds 1 rad — the trajectories have fully decorrelated. This is expected: the divergence rate matches the Lyapunov timescale from Figure 5, and the short-time agreement confirms that the RK4 implementation is correct.

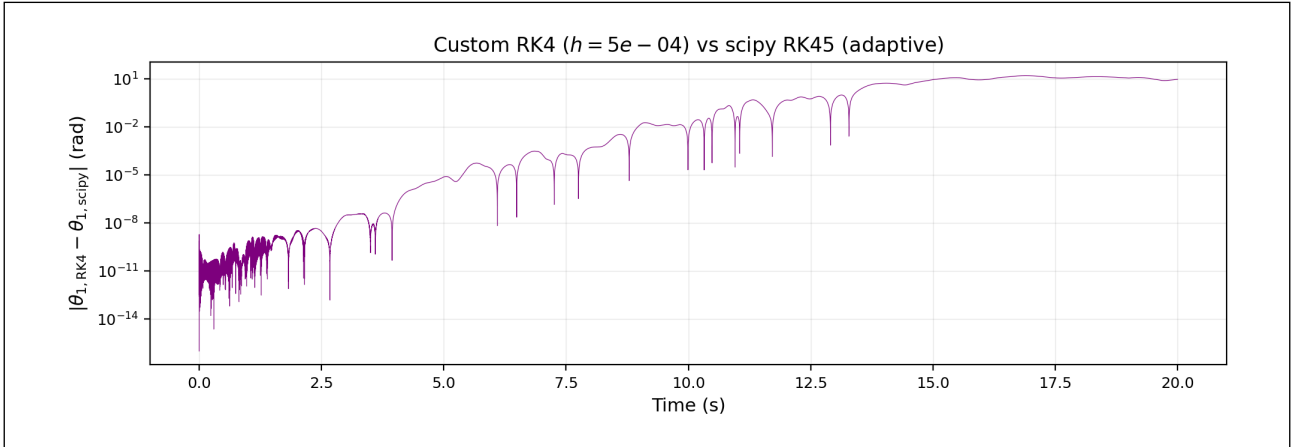


Figure 6: Absolute difference in θ_1 between custom RK4 ($h = 5 \times 10^{-4}$ s) and scipy RK45 (adaptive, $\text{rtol} = 10^{-10}$). Short-time agreement at 10^{-11} diverges chaotically after ≈ 5 s.

5.7 Bifurcation Diagram

Figure 7 maps the long-time behaviour as τ_0 is swept from 0.5 to 10.0 N · m. At each amplitude the system is integrated for 40 s of transient followed by 30 s of Poincaré sampling at the driving period; the sampled angular velocities $\dot{\theta}_i$ (rather than wrapped angles) are plotted. For $\tau_0 < 2$ N · m the Poincaré points cluster tightly near zero — the damping dominates and the system responds periodically. As τ_0 increases past ≈ 2 –3 N · m, the clusters broaden into scattered bands: the system has entered chaos. At higher amplitudes the scatter fills an increasingly wide range of angular velocities. The transition is gradual, consistent with a Feigenbaum route to chaos.

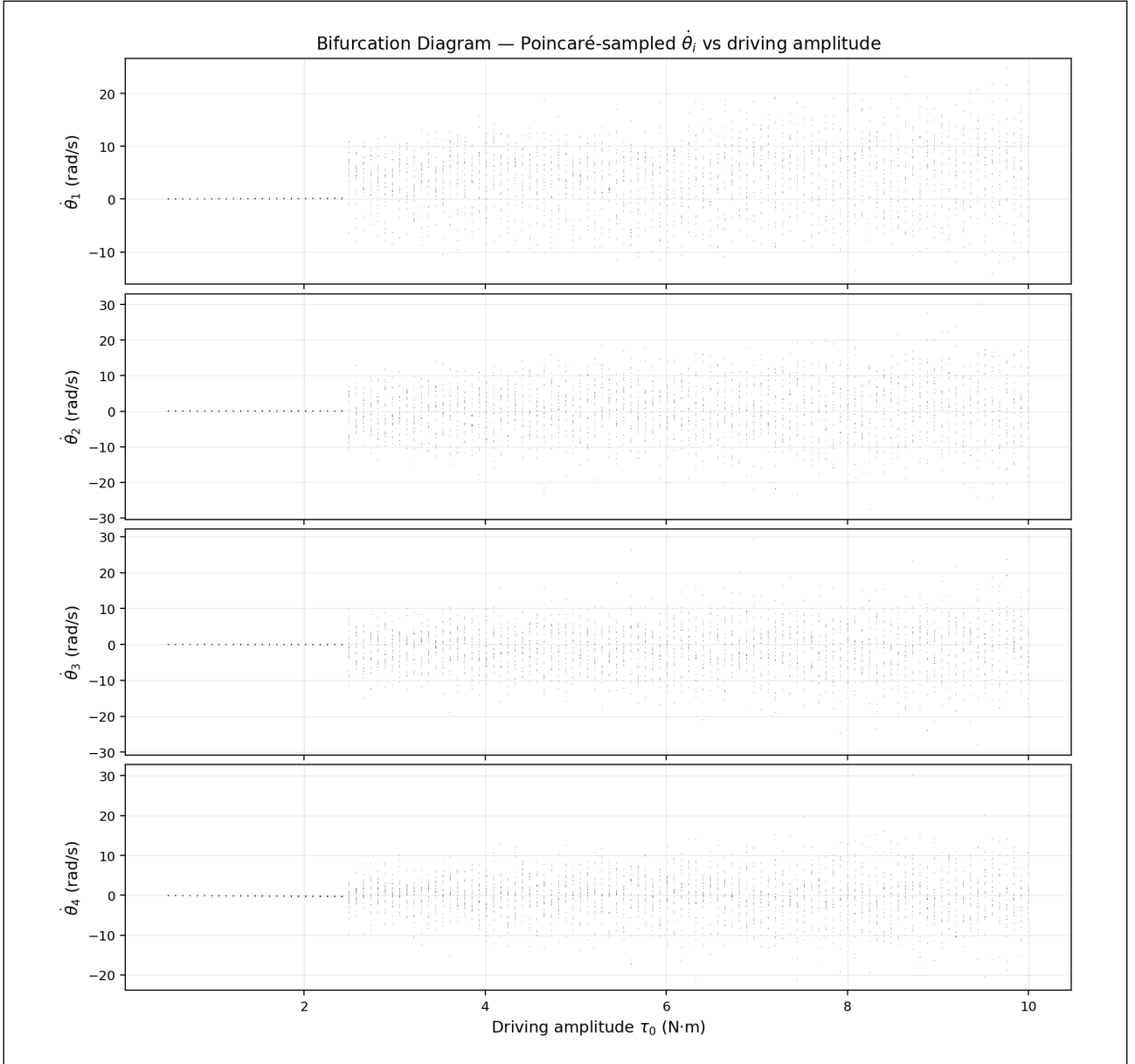


Figure 7: Bifurcation diagram: Poincaré-sampled angular velocities $\dot{\theta}_i$ as a function of driving amplitude τ_0 . The transition from periodic clustering to chaotic scattering occurs near $\tau_0 \approx 2\text{--}3 \text{ N} \cdot \text{m}$.

5.8 Animation

An animated GIF ([plots/animation.gif](#)) shows the quadruple pendulum swinging for the first 15 s at 30 fps, with a trailing path for the tip of the fourth bob. The animation was generated entirely within the script using `matplotlib.animation.FuncAnimation` and `PillowWriter`.

6 Reflections and Discussion

6.1 Physical Insights

The simulation reveals three features that are difficult to appreciate from the equations alone:

- **The competition between driving and damping sets the dynamics.** Without damping, the system accumulates energy indefinitely and blows up numerically. With damping, the injected and dissipated power balance on average, producing a bounded but chaotic steady state. The damping coefficient $b = 0.10$ is light enough to permit chaos at $\tau_0 = 4.0 \text{ N} \cdot \text{m}$ but strong enough to keep all trajectories bounded over arbitrarily long integrations.

- **Sensitivity to initial conditions is practical, not just theoretical.** Even with double-precision arithmetic (≈ 16 significant digits), two trajectories starting 10^{-6} rad apart are completely decorrelated within about 8 s. The `scipy` comparison tells the same story: two correct integrators with slightly different truncation errors diverge just as fast as two perturbed initial conditions.
- **The transition to chaos is gradual.** The bifurcation diagram shows that the system does not jump abruptly from order to chaos. Below $\tau_0 \approx 2 \text{ N} \cdot \text{m}$, damping wins and the response is periodic. Above $\approx 3 \text{ N} \cdot \text{m}$ the Poincaré samples scatter broadly. The four degrees of freedom provide enough channels for instability that the chaotic onset occurs at a relatively low driving-to-damping ratio.

6.2 Limitations

The model assumes rigid, massless rods and point masses. A physical quadruple pendulum would have distributed mass and finite rod stiffness. The damping model ($-b\dot{\theta}_i$) is the simplest possible: proportional to absolute angular velocity rather than to the relative velocity at each joint. A more realistic friction model ($-b(\dot{\theta}_i - \dot{\theta}_{i-1})$) would change the quantitative bifurcation thresholds but not the qualitative phenomena.

The fixed time step ($\Delta t = 5 \times 10^{-4} \text{ s}$) handles the default parameters well, but at very high driving amplitudes the outer pendulums whip fast enough that even this step size may accumulate error. An adaptive step-size controller (as in `scipy`'s RK45) would be more robust, but the fixed-step RK4 was chosen to satisfy the project requirement of implementing one's own algorithm.

7 Conclusion

The driven, damped quadruple pendulum was simulated by deriving the equations of motion from the Lagrangian (including viscous damping), recasting them as eight first-order ODEs, and integrating with a hand-coded RK4 scheme. The 4×4 mass-matrix system is solved at each RK4 stage by LU decomposition. The simulation reproduces the expected phenomenology of a chaotic driven-dissipative system: aperiodic time series with full rotations, space-filling phase portraits, scattered Poincaré sections, exponential sensitivity to initial conditions over 8 orders of magnitude, and a bifurcation diagram showing the transition from periodic to chaotic response as driving amplitude increases.

8 References

1. J. M. T. Thompson and H. B. Stewart, *Nonlinear Dynamics and Chaos*, 2nd ed. (Wiley, 2002).
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3. W. H. Press, S. A. Teukolsky, W. T. Vetterling, and B. P. Flannery, *Numerical Recipes*, 3rd ed. (Cambridge University Press, 2007).
4. PC3236 Computational Methods in Physics, Project Handout, NUS (2025/26).

9 Declaration on the Use of Generative AI

I declare that I **HAVE** used generative AI tools to produce this assignment.

I acknowledge that generative AI was used in the following manner:

AI Tool Used	My Prompt and AI Output	How the Output Was Used
Claude	Prompt: “Help me structure the Python simulation for the driven quadruple pendulum, including the Lagrangian derivation, RK4 integrator, and plotting routines.” Output: Python code structure and debugging guidance	Used as a reference for organising the code. All equations were derived by me and verified against textbook sources. The RK4 algorithm was implemented from my lecture notes. I manually corrected sign errors found by energy-conservation testing.
Claude	Prompt: “Help with Typst formatting for figures, tables, equations, and the code-to-equation mapping table.” Output: Typst formatting suggestions	Used only for formatting. All physical analysis, interpretation, and report content were written and verified by me.

Table 2: AI Tool Usage Declaration